# *36 Testing ASP.NET Core applications*

## This chapter covers

- Writing unit tests for custom middleware, API controllers, and minimal API endpoints
- Using the Test Host package to write integration tests
- Testing your real application's behavior with WebApplicationFactory
- Testing code dependent on Entity Framework Core with the in-memory database provider

In chapter 35 I described how to test .NET 7 applications using the xUnit test project and the .NET Test software development kit (SDK). You learned how to create a test project, add a project reference to your application, and write unit tests for services in your app.

In this chapter we focus on testing ASP.NET Core applications specifically. In sections 36.1 and 36.2 we'll look at how to test common features of your ASP.NET Core apps: custom middleware, API controllers, and minimal API endpoints. I show you how to write isolated unit tests for both, much like you would any other service, and I'll point out the tripping points to watch for.

To ensure that components work correctly, it's important to test them in isolation. But you also need to test that they work correctly in a middleware pipeline. ASP.NET Core provides a handy Test Host package that lets you easily write these integration *tests* for your components. You can even go one step further with the `WebApplicationFactory` helper class and test that your app is working correctly. In section 36.3 you'll see how to